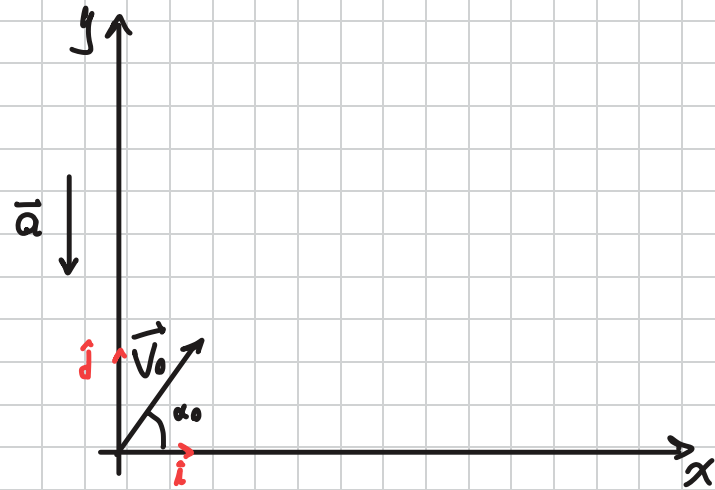


Week 3

Projectile motion



$$a_x = 0, a_y = -g$$

initial position $x_0 = 0, y_0 = 0$

initial velocity
$$\vec{V}_0 = v_0 \cos \alpha_0 \hat{i} + v_0 \sin \alpha_0 \hat{j}$$

X: $a_x = 0 \Rightarrow$ Constant velocity

$$x = x_0 + v_{0x} t = v_0 \cos \alpha_0 t \quad (1)$$

Y: Constant acceleration $a_y = -g$

$$v_y = v_{0y} + a_y t = v_0 \sin \alpha_0 - gt \quad (2)$$

$$y = y_0 + v_{0y} t + \frac{1}{2} a_y t^2 = v_0 \sin \alpha_0 t - \frac{1}{2} g t^2 \quad (3)$$

Solve for t from eqn (1)

$$t = \frac{x}{v_0 \cos \alpha_0} \quad \text{Sub into (3)}$$

$$y = \cancel{V_0} \sin \alpha_0 \frac{x}{\cancel{V_0} \cos \alpha_0} - \frac{1}{2} g \frac{x^2}{V_0^2 \cos^2 \alpha_0}$$

$$y = \tan \alpha_0 x - \frac{g}{2V_0^2 \cos^2 \alpha_0} x^2$$

Equation of the form $y = ax - bx^2$
has a graph in the shape of a parabola

Require that $y=0$

$$\tan \alpha_0 x - \frac{g}{2V_0^2 \cos^2 \alpha_0} x^2 = 0$$

$$x \left(\tan \alpha_0 - \frac{g}{2V_0^2 \cos^2 \alpha_0} x \right) = 0$$

One solution is $x=0$ (launch point)

the other: $\tan \alpha = \frac{g}{2V_0^2 \cos^2 \alpha} X$

$$X = \frac{2V_0^2 \cos^2 \alpha_0 \tan \alpha_0}{g} = \frac{2V_0^2 \sin \alpha_0 \cos \alpha_0}{g}$$

or $x_{\max} = \frac{V_0^2}{g} \sin 2\alpha_0$

For maximum distance traveled (horizontally)

we must have $\sin 2\alpha_0 = 1$

$$\Rightarrow 2\alpha_0 = \frac{\pi}{2} \Rightarrow \alpha_0 = \frac{\pi}{4}$$

optimize launch
angle

What is the
maximum height?

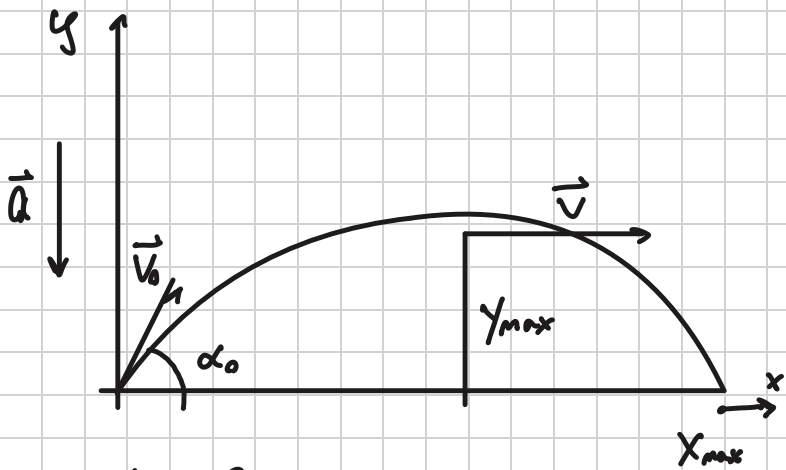
At maximum height, $V_y = 0$

$$V_y^2 = V_{0y}^2 + 2ay \Rightarrow V_y = 0, a = -g$$

$$V_{0y} = V_0 \sin \alpha_0$$

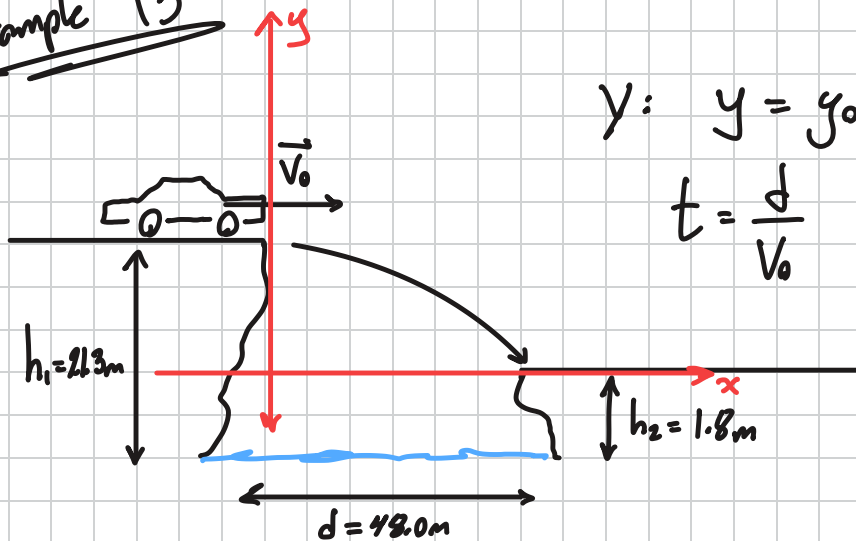
$$0 = V_0^2 \sin^2 \alpha_0 - 2g y_{\max} \Rightarrow$$

$$y_{\max} = \frac{V_0^2 \sin^2 \alpha_0}{2g}$$



$$y_0 = h_1 - h_2$$

Example 13



$$x: x = v_0 t$$

$$y: y = y_0 + v_{0y}t + \frac{1}{2}at^2 = (h_1 - h_2) - \frac{1}{2}gt^2$$

$$t = \frac{d}{v_0}$$

time for y to drop to zero:

$$(h_1 - h_2) - \frac{1}{2}gt^2 = 0$$

$$t^2 = \frac{2(h_1 - h_2)}{g}$$

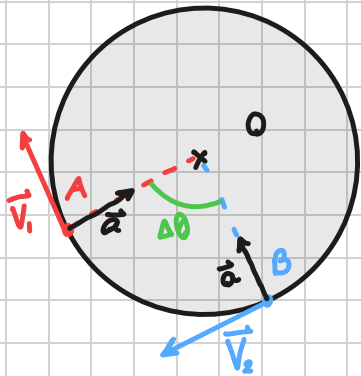
Set the two equal

$$t = \sqrt{\frac{2(h_1 - h_2)}{g}}$$

$$\frac{d}{V_0} = \sqrt{\frac{2(h_1 - h_2)}{g}} \Rightarrow V_0 = d \sqrt{\frac{g}{2(h_1 - h_2)}}$$

Circular motion

① Constant speed



$$v = |\vec{v}_1| + |\vec{v}_2| = \text{constant}$$

$$\Delta\theta = \frac{\Delta s}{R}, \quad \Delta s = \text{arc length AB}$$

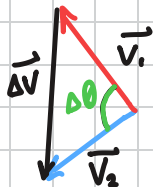
$$\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$$

$$|\vec{a}| = \frac{|\Delta\vec{v}|}{\Delta t} = a$$

$$a = \frac{|\Delta\vec{v}|}{\Delta t} \approx \frac{v \Delta\theta}{\Delta t} = v \cdot \frac{\Delta s}{R} \cdot \frac{1}{\Delta t} = \frac{v}{R} \cdot \frac{\Delta s}{\Delta t} = v$$

$$\Rightarrow \boxed{a = \frac{v^2}{R}}$$

$\vec{a} \perp \vec{v}$ at all times, points to center of circle



if $\Delta\theta$ is small

$$|\Delta\vec{v}| \approx |\vec{v}| \cdot \Delta\theta = v \Delta\theta$$

Centripetal acceleration

Period. T (time to go around circle once)

$$VT = 2\pi r \Rightarrow V = \frac{2\pi r}{T} \Rightarrow \boxed{V = \omega r}$$

$\frac{2\pi}{T} = \omega$ angular frequency or angular velocity

$$a = \frac{V^2}{r} = \frac{4\pi^2 r^2}{T^2} \cdot \frac{1}{r} = \frac{4\pi^2 r}{T^2}$$

$$\text{or } a = \frac{\omega^2 r^2}{r} \Rightarrow \boxed{a = \omega^2 r}$$

② Non-uniform motion

V is not constant

$$\vec{a} = \vec{a}_{\text{rad}} + \vec{a}_{\text{tan}}$$

$$\begin{array}{l} \vec{a}_{\text{rad}} \perp \vec{v} \\ \vec{a}_{\text{tan}} \parallel \vec{v} \end{array}$$

$$\vec{a}_{\text{rad}} = \frac{v^2}{r}$$

$$\vec{a}_{\text{tan}} = \frac{dv}{dt} \rightarrow \text{rate of change of the speed}$$

Example

3.28 Helicopter has blades of

length $L = 3.40\text{m}$ rotates at 550 rev/min

What is v at blade tip? what is the $\vec{a}_{\text{rad}}$
as a multiple of g ?



What is the angular velocity ω ?

$$\begin{aligned}\omega &= 550 \frac{\cancel{\text{rev}}}{\cancel{\text{min}}} \cdot 2\pi \frac{\text{rad}}{\cancel{\text{rev}}} \cdot \frac{1}{60} \frac{\cancel{\text{min}}}{\text{s}} \\&= \frac{550 \cdot 2\pi}{60} \frac{\text{rad}}{\text{s}} \\&= \frac{55\pi}{3} \frac{\text{rad}}{\text{s}}\end{aligned}$$

$$v = \omega \cdot L = \frac{55\pi}{3} \cdot 3.40 \frac{\text{m}}{\text{s}} = 1958 \text{ m/s} \approx 196 \text{ m/s}$$

$$a = \omega^2 r = \omega L = \dots$$

$$\Rightarrow \frac{a}{g} = \dots \quad \underline{\text{exercise}}$$

3.30 Example

V, T

Calculate a in terms of V, T but not r

$$Q = \frac{V^2}{r}, \text{ but } V \cdot T = 2\pi r$$

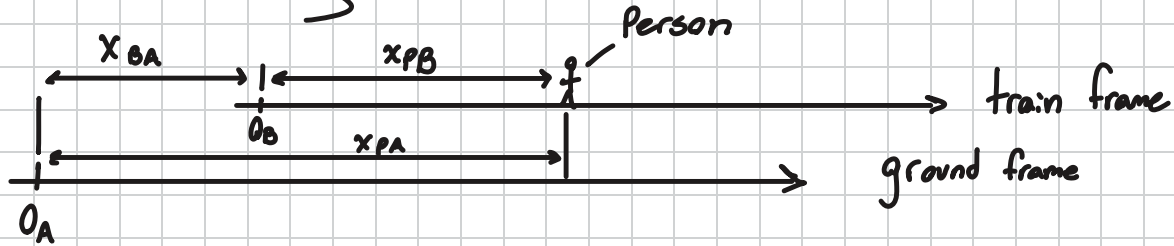
$$\Rightarrow r = \frac{V \cdot T}{2\pi}$$

$$Q = \frac{\cancel{V^2}}{\frac{\cancel{V} \cdot T}{2\pi}} \Rightarrow \boxed{a = 2\pi \cdot \frac{V}{T}}$$

Q if V doubles, by what factor must T change so that a remains the same?



relative velocity



$$x_{PA} = x_{PB} + x_{BA} \quad \text{valid at all times}$$

Apply $\frac{d}{dt}$ on the previous equation

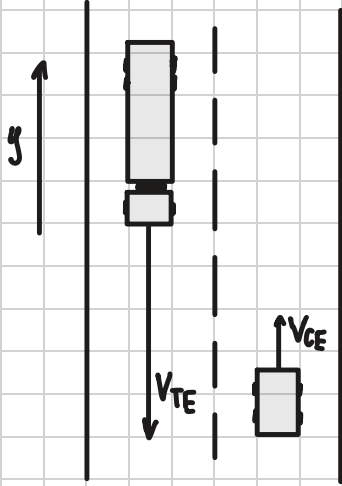
$$\frac{dx_{PA}}{dt} = \frac{dx_{PB}}{dt} + \frac{dx_{BA}}{dt}$$

$$V_{PA} = V_{PB} + V_{BA}$$

Note: in 3D (or 2D)
it becomes a vector
equation

Special case: $P = A$

$$0 = V_{AA} = V_{AB} + V_{BA} \Rightarrow V_{AB} = -V_{BA}$$



C - car

E - earth

T - truck

$$V_{CE} = 88 \text{ km/h}$$

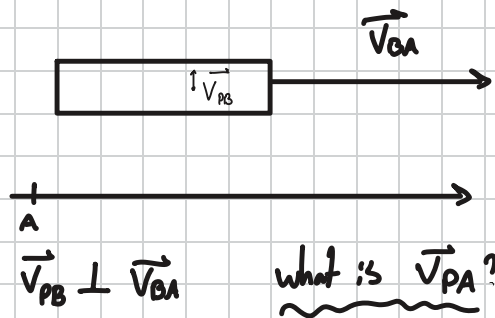
$$V_{TE} = -104 \text{ km/h}$$

$$\begin{aligned} V_{TC} &= V_{TE} + V_{EC} \\ &= V_{TE} - V_{CE} \\ &= -104 \text{ km/h} - 88 \text{ km/h} \\ &= -192 \text{ km/h} \end{aligned}$$

What is the relative velocity of the truck to the car?

2D case

$$\vec{V}_{PA} = \vec{V}_{PB} + \vec{V}_{BA}$$



$$|\vec{V}_{BA}| = 3.0 \text{ m/s}$$

$$|\vec{V}_{PB}| = 1.0 \text{ m/s}$$

$$\vec{V}_{PA} = \vec{V}_{PB} + \vec{V}_{BA}$$



$$V_{PA} = \sqrt{V_{BA}^2 + V_{PB}^2} = \sqrt{9.0 + 1.0} \text{ m/s}$$

$$V_{PA} = 3.2 \text{ m/s}$$

$$\tan \phi = \frac{V_{PB}}{V_{BA}} = \frac{1.0}{3.0} \Rightarrow \phi = 18^\circ$$

Forces and inertia

word of the day
perturb

Material objects are subject to interactions that perturb their motion

Example

Sun - Earth system is
subject to gravitational interaction

Force is a measure of interaction

① contact forces (forces between objects that come in contact with each other)

- normal forces
- friction forces

② long range forces $\rightarrow$ fields

- gravity

- electro magnetic forces